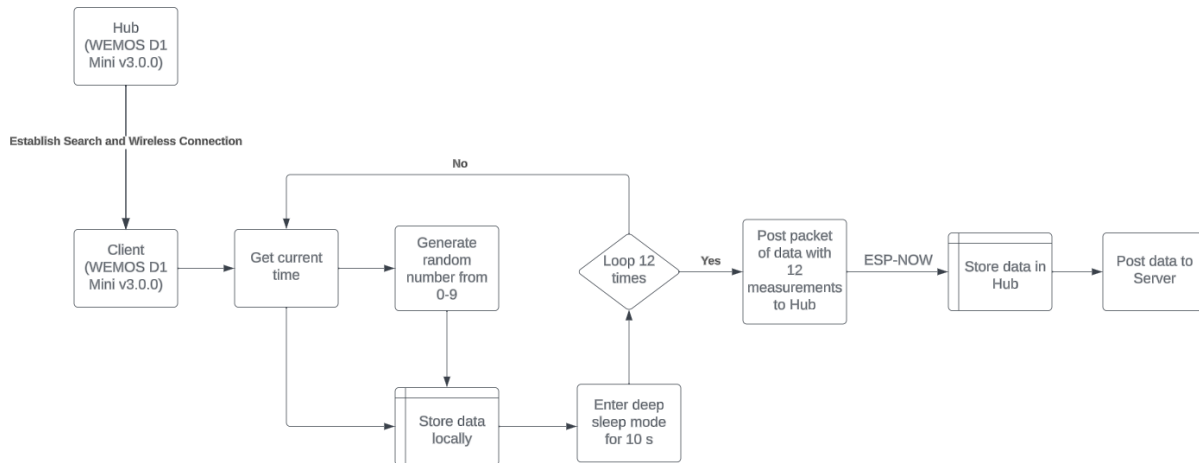


Statement of Work

Overview:

The goal of this project is to implement a data transfer system between two microcontrollers and a remote server. The system architecture is diagramed below:



The process is shown in the system architecture, but will be written out here for clarity:

1. The Hub board (WEMOS D1 Mini v3.0.0) upon powering on will default to a 'Search Mode' and pair to an available Client board (WEMOS D1 Mini v3.0.0) on the same local area network.
2. After connection is established between the two boards, the Client board begins in deep sleep mode.
3. Client gets current time and generates a random number from 0 to 9 inclusive.
4. Data is formatted and saved locally on the Client.
5. Client enters deep sleep mode for 10 seconds.
6. 1-4 are repeated 12 times such that 12 rows of data are saved on the Client.
7. All data on the Client is then formatted as a packet.
8. Data is then posted from Client to Hub (WEMOS D1 MINI v3.0.0) via ESP-NOW.
9. Data is saved locally on Hub.
10. Hub is posted to a remote server via SSH to a Raspberry Pi out of network.

Objectives:

Objective 1: Establish automatic communication between Hub and Client.

Please complete firmware code that allows the Hub to search for the (powered on) Client board and establish a connection. This is equivalent to Process 1 in the Overview.

Objective 2: Generate and store data locally on Client.

Please complete firmware code that allows for collecting data from deep sleep mode on a WEMOS D1 Mini v3.0.0 board in a loop. This is equivalent to Processes 2-6 outlined in the Overview.

Objective 3: Post data from Client to Hub.

Format the 12 rows of data (if necessary) and post it to the Hub via ESP-NOW. This is equivalent to Processes 7-9 outlined in the Overview.

Objective 4: Post data from the Hub to a remote server.

The data will need to be accessed remotely and will thus need to be posted to a Raspberry Pi server on a different WiFi network. Assume that the server can be accessed via SSH and data can be saved in the home directory. This is equivalent to Process 10 in the Overview.

Deliverable:

Firmware code (Arduino or similar) for both the Client and Hub that completes Objectives 1-4.